

Applications on Differential Equations

Application 1: Cooling/Warming law

the mathematical formulation of Newton's empirical law of cooling of an object is given by the linear first-order differential equation

$$\frac{dT}{dt} = \alpha(T - T_m)$$

Where T is the temperature of the object, T_m the surrounding temperature and α is constant

Example 4.1: When a chicken is removed from an oven, its temperature is measured at 300°F. Three minutes later its temperature is 200°F. How long will it take for the chicken to cool off to 100°F at room temperature of 70°F.

Solution:

$$\frac{dT}{dt} = \alpha(T - 70)$$

$$\frac{dT}{T - 70} = \alpha dt \rightarrow \int \frac{dT}{T - 70} = \int \alpha dt$$

$$\ln(T - 70) = \alpha t + C \rightarrow T = 70 + ke^{\alpha t}$$

$$\text{at } t = 0 \quad T = 300 \rightarrow k = 230,$$

$$\text{at } t = 3 \quad T = 200 \rightarrow \alpha = -0.19018$$

$$T(t) = 70 + 230e^{-0.19018t}$$

Put $T = 100$, get $t = 10.7 \text{ min.}$

Application 2:- POPULATION DYNAMICS:

Thomas Malthus (1798):-

Assumption:- The population growth rate at any time (human of a country for example) is proportional to the total population at that time.

State Variable:- $P(t)$ the population at time t

Differential Equation:-

$$\frac{dP(t)}{dt} \propto P(t)$$
$$\frac{dP(t)}{dt} = kP(t)$$

Note:- This model fails later on since many other factors are not taken into consideration such as resources, medical, immigration, But it is still used for small populations such as Bacteria growth in the lab.

EXAMPLE 1 Bacterial Growth

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

The governing DE:-

$$\frac{dP}{dt} - kP = 0$$

The general solutions of the DE (HOW??)

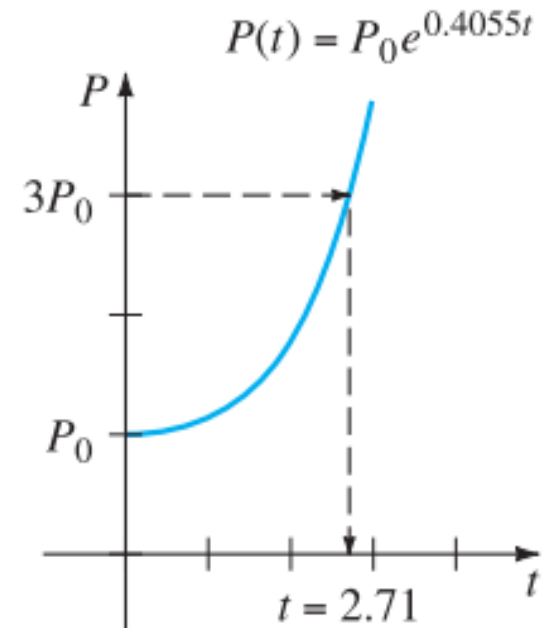
$$P(t) = ce^{kt}$$

The specific solution of this problem (at $t = 1$, $P(t) = 1.5P_0$):-

$$P(t) = P_0 e^{0.4055t}$$

The time when $P(t) = 3P_0$ is:-

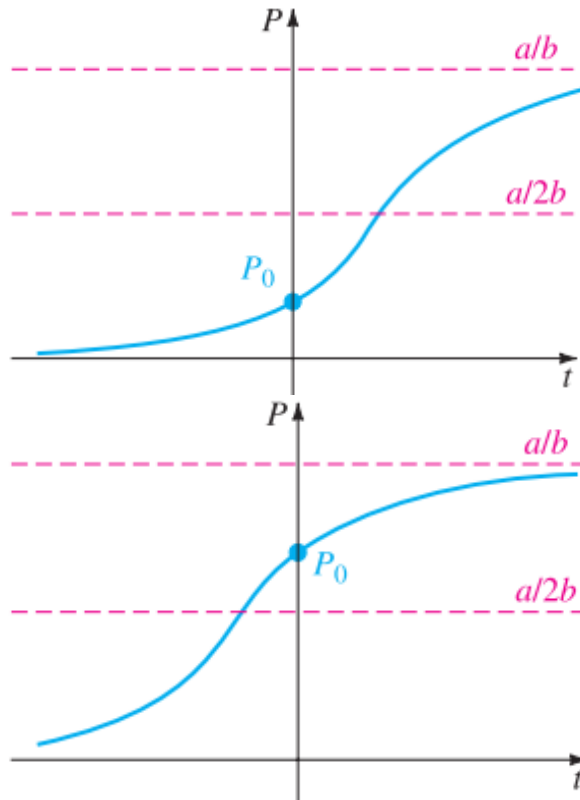
$$t = \frac{\ln 3}{0.4055} \approx 2.71 \text{ h}$$



LOGISTIC EQUATION (Non-linear Population Model, Belgian mathematician-biologist P. F. Verhulst 1840)

$$\frac{dP}{dt} = P(a - bP).$$

SOLUTION OF THE LOGISTIC EQUATION



$$\left(\frac{1/a}{P} + \frac{b/a}{a - bP} \right) dP = dt$$

$$\frac{1}{a} \ln|P| - \frac{1}{a} \ln|a - bP| = t + c$$

$$\ln \left| \frac{P}{a - bP} \right| = at + ac$$

$$\frac{P}{a - bP} = c_1 e^{at}.$$

$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}$$

Problem:- Solve the Gompertz differential equation (growth of solid tumors)

$$\frac{dP}{dt} = P(a - b \ln P)$$

Application 3: DRAINING A TANK:

Bernoulli's principle (1738):- Incompressible flow equation at any point:

$$\frac{v^2}{2} + gh + \frac{p}{\rho} = \text{constant}$$

Where v, g, h, p , and ρ are the fluid flow speed, acceleration due to gravity, the height of this point measured from a certain reference, pressure at that point, and density of the fluid.

$$\left(\frac{v^2}{2} + gh + \frac{p}{\rho} \right)_{\text{top}} = \left(\frac{v^2}{2} + gh + \frac{p}{\rho} \right)_{\text{bottom}}$$

$$v(t) = \sqrt{2gh(t)}$$

State Variable:- $h(t)$ the height

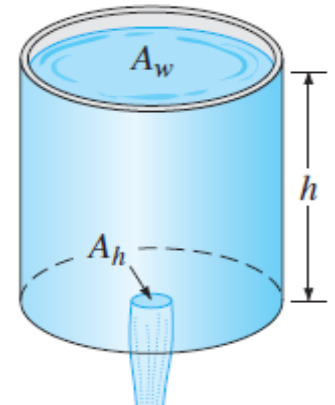
Assumption:- Big tank (neglect the speed of fluid at the upper surface)

Differential Equation:- Volume = Area*Height

$$V(t) = A_w h \Rightarrow \frac{dV(t)}{dt} = A_w \frac{dh}{dt}$$

$$\frac{dV(t)}{dt} = -A_h v(t) = -A_h \sqrt{2gh}$$

$$\frac{dh(t)}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}$$



$$\frac{dh(t)}{dt} = -\frac{A_h}{A_w} \sqrt{2gh},$$

$$\frac{dh}{\sqrt{h}} = -\frac{A_h}{A_w} \sqrt{2g} dt$$

By integrating both sides:-

$$\sqrt{h} = -\frac{A_h}{A_w} \sqrt{0.5g} t + c_1 = c_1 - kt, \quad k = \frac{A_h}{A_w} \sqrt{0.5g}$$

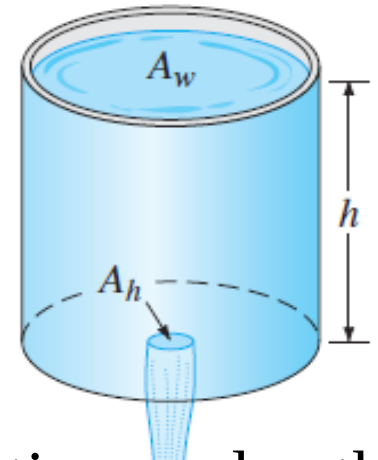
If the initial condition is $h(t = 0) = H_0$, then the value of c_1

$$c_1 = \sqrt{H_0}$$

Then the height as a function of the time is given by:-

$$h(t) = (\sqrt{H_0} - kt)^2$$

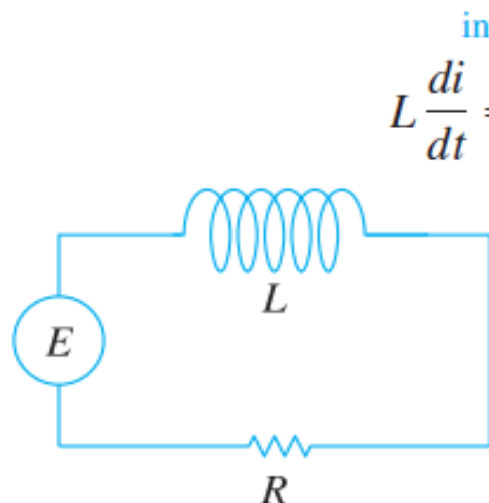
The tank will be empty when $h(t) = 0$ at $t = \sqrt{H_0}/k$



More Ideas?? What will be the DE and the solution under the following conditions:

- 1- the hole is 1m above the base??
- 2- there are many holes in the same levels
- 3- Two holes on different levels

Application 4: Series Circuits: $v(t)$:voltage



$$L \frac{di}{dt} + Ri = E(t)$$

This is Linear 1st order DE, then

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E(t)}{L}$$

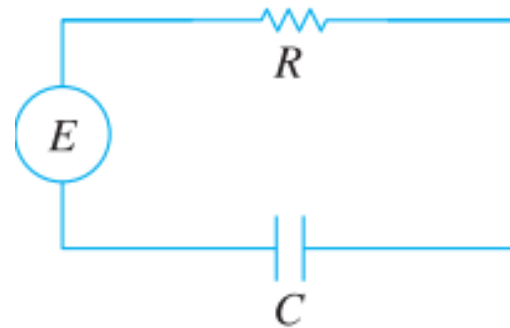
$$\mu(t) = e^{\frac{R}{L}t}$$

$$i(t) = e^{-\frac{R}{L}t} \left(\int e^{\frac{R}{L}t} \frac{E(t)}{L} dt + c \right)$$

Given the values of R , L , and $E(t)$, the function $i(t)$ will be determined.

inductor resistor capacitor

$$L \frac{di}{dt} = L \frac{d^2q}{dt^2}, \quad iR = R \frac{dq}{dt}, \quad \text{and} \quad \frac{1}{C}q$$



$$R \frac{dq}{dt} + \frac{1}{C}q = E(t)$$

This is Linear 1st order DE, then

$$\frac{dq}{dt} + \frac{1}{RC}q = \frac{E(t)}{R}$$

$$\mu(t) = e^{\frac{1}{RC}t}$$

$$q(t) = e^{-\frac{1}{RC}t} \left(\int e^{\frac{1}{RC}t} \frac{E(t)}{R} dt + c \right)$$

Given the values of R , C , and $E(t)$, the function $q(t)$ will be determined.

Application 5:- Radioactive Decay

Introduction:-

The nucleus of an atom $\rightarrow$ protons + neutrons

Unstable protons and neutrons $\rightarrow$ the atoms transform into atoms of another substance $\rightarrow$ radioactive nuclei.

State Variable:- $A(t)$ the number of nuclei

Assumption:- The rate of change of $A(t)$ is proportional to $A(t)$

Differential Equation:-

$$\frac{dA(t)}{dt} \propto A(t)$$
$$\frac{dA(t)}{dt} = kA(t)$$

Note:- k should be negative in this case (decaying behavior).

EXAMPLE 2 Half-Life of Plutonium

A breeder reactor converts relatively stable uranium 238 into the isotope plutonium 239. After 15 years it is determined that 0.043% of the initial amount A_0 of plutonium has disintegrated. Find the half-life of this isotope if the rate of disintegration is proportional to the amount remaining.

The governing DE:-

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

The general solutions of the DE (HOW??)

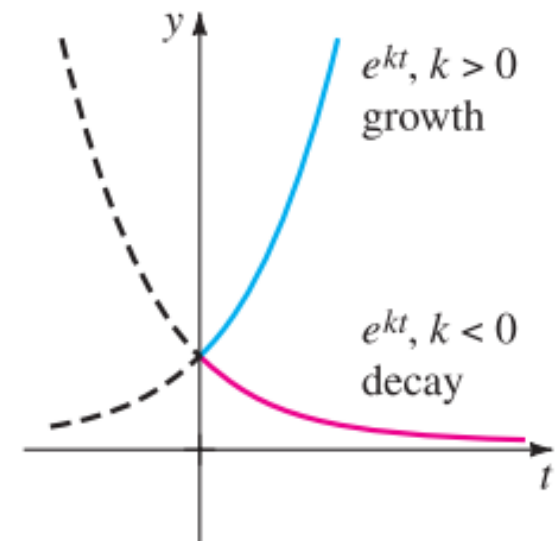
$$A(t) = A_0 e^{kt}.$$

The specific solution of this problem:-

$$0.99957A_0 = A_0 e^{15k}, \quad A(t) = A_0 e^{-0.00002867t}$$

The time when $A(t) = 0.5A_0$ is:-

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ yr}$$



Part 2:

Numerical Solutions Of First Order ODEs

Numerical Solutions Of First Order ODEs

- General Form

$$y' = f(x, y) , \quad y(x_0) = y_0 \quad \textit{Evaluta } y(x_f)$$

$$\text{e.g. } y' = \tan^{-1}\left(\frac{x}{x^2+y}\right) , y(1) = 3 \quad \textit{Evaluta } y(2)$$

Euler's Method:

Assume $y(x)$, then

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$$y'(x) = \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{(x+h) - (x)}$$

$$y'(x) = \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$$

If $h \llllll x$

$$y'(x) \cong \frac{y(x+h) - y(x)}{h}$$

$$y(x+h) \cong y(x) + hy'(x) \quad (1)$$

$$y(x + h) \cong y(x) + hy'(x) \quad (1)$$

e.g. $y' = x^2\sqrt{y}$ $y(3) = 5$, Evaluate $y(3.1)$

Using (1) with $h = 0.01$

$$y(3.01) \cong y(3) + (0.01)y'(3)$$

$$y(3.01) \cong 5 + (0.01) \left((3)^2\sqrt{5} \right) = 5.20125$$

Using (1)

$$y(3.02) \cong y(3.01) + (0.01)y'(3.01)$$

$$\begin{aligned} y(3.02) &\cong 5.20125 + (0.01) \left((3.01)^2\sqrt{5.20125} \right) \\ &= 5.40788 \end{aligned}$$

Similarly, we can obtain

$y(3.03), y(3.04), \dots \dots \dots, y(3.1)$ (In total 10 steps)

EULER'S METHOD

$$y' = f(x, y), \quad y(x_0) = y_0$$

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$$y_1 = y_0 + hf(x_0, y_0)$$

$$y_2 = y_1 + hf(x_1, y_1)$$

EXAMPLE 1 Euler's Method

Consider the initial-value problem $y' = 0.1\sqrt{y} + 0.4x^2$, $y(2) = 4$. Use Euler's method to obtain an approximation of $y(2.5)$ using first $h = 0.1$ and then $h = 0.05$.

$$y_{n+1} = y_n + h(0.1\sqrt{y_n} + 0.4x_n^2)$$

TABLE 2.1 $h = 0.1$

x_n	y_n
2.00	4.0000
2.10	4.1800
2.20	4.3768
2.30	4.5914
2.40	4.8244
2.50	5.0768

Matlab Code

EXAMPLE 1 Euler's Method

Consider the initial-value problem $y' = 0.1\sqrt{y} + 0.4x^2$, $y(2) = 4$. Use Euler's method to obtain an approximation of $y(2.5)$ using first $h = 0.1$ and then $h = 0.05$.

```
xi=2;
xf=2.5;
x(1)=xi;
y(1)=4.0;
h=0.05;
jmax=(xf-xi)/h;
for j=1:jmax
    x(j+1)=x(j)+h;
    y(j+1)=y(j)+h*(0.1*sqrt(y(j))+0.4*(x(j))^2);
end
hold on;
plot(x,y,'b')
```

TABLE 2.2 $h = 0.05$

x_n	y_n
2.00	4.0000
2.05	4.0900
2.10	4.1842
2.15	4.2826
2.20	4.3854
2.25	4.4927
2.30	4.6045
2.35	4.7210
2.40	4.8423
2.45	4.9686
2.50	5.0997

Example:- solve $y' = xy + y, y(0) = 2$ analytically then get the first two steps using Euler method with $h = 0.01$ and their errors.

Analytical Solution:- By separation $y' = y(x + 1)$

$$\frac{dy}{y} = (x + 1)dx \quad \Rightarrow \ln y = \frac{x^2}{2} + x + c$$

$$\Rightarrow y = e^{\frac{x^2}{2} + x + c} = c_1 e^{\frac{x^2}{2} + x} \quad \text{at } x = 0, y = 2 \Rightarrow c_1 = 2$$

$$y(x) = 2e^{\frac{x^2}{2} + x}$$

Euler Method:-

$$y_{n+1} = y_n + hf(x_n, y_n) = y_n + h(y_n(x_n + 1))$$

$$n = 0 \quad \Rightarrow x_0 = 0, y_0 = 2 \Rightarrow$$

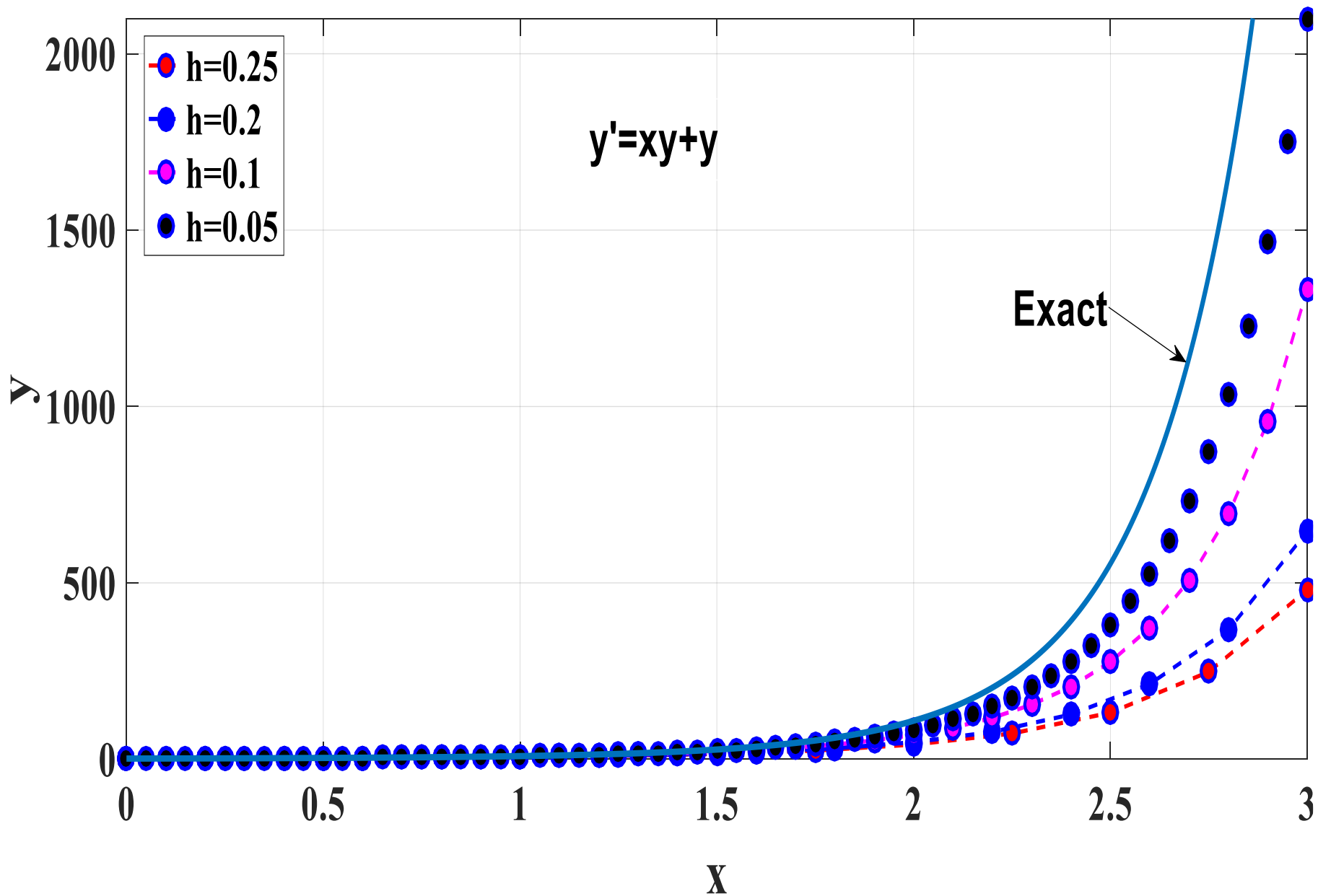
$$y_1 = y(0.01) = 2 + 0.01(2(0 + 1)) = 2.02$$

$$n = 1 \quad \Rightarrow x_1 = 0.01, y_1 = 2.02 \Rightarrow$$

$$y_2 = y(0.02) = 2.02 + 0.01(2.02(0.01 + 1)) = 2.040402$$

$$\text{Error(at } x = 0.01) = 2e^{0.01005} - 2.02 = 2.0134 \times 10^{-4}$$

$$\text{Error(at } x = 0.02) = 2e^{0.0202} - 2.040402 = 3.908 \times 10^{-4}$$



Euler's Method

EXAMPLE 2.1. Find approximate to $x(1)$ if $x'(t) = t^2 - e^x \sin(t)$, $x(0) = 1$ with $h = 0.25$.

Solution Here the interval is $[0, 1]$, so $a = 0, b = 1$. Since $h = \frac{b-a}{n} = 0.25$ we have $n = 4$ and we need to compute x_1, x_2, x_3, x_4 starting with $x_0 = x(0) = 1$.

$$x_{n+1} = x_n + h f(x_n, t_n) = x_n + h(t_n^2 - e^{x_n} \sin(t_n)), t_n = t_0 + nh$$

$n=0 \rightarrow x_0 = 1, t_0 = 0$

$$\rightarrow x_1 = 1 + 0.25(0^2 - e^1(0)) = 1$$

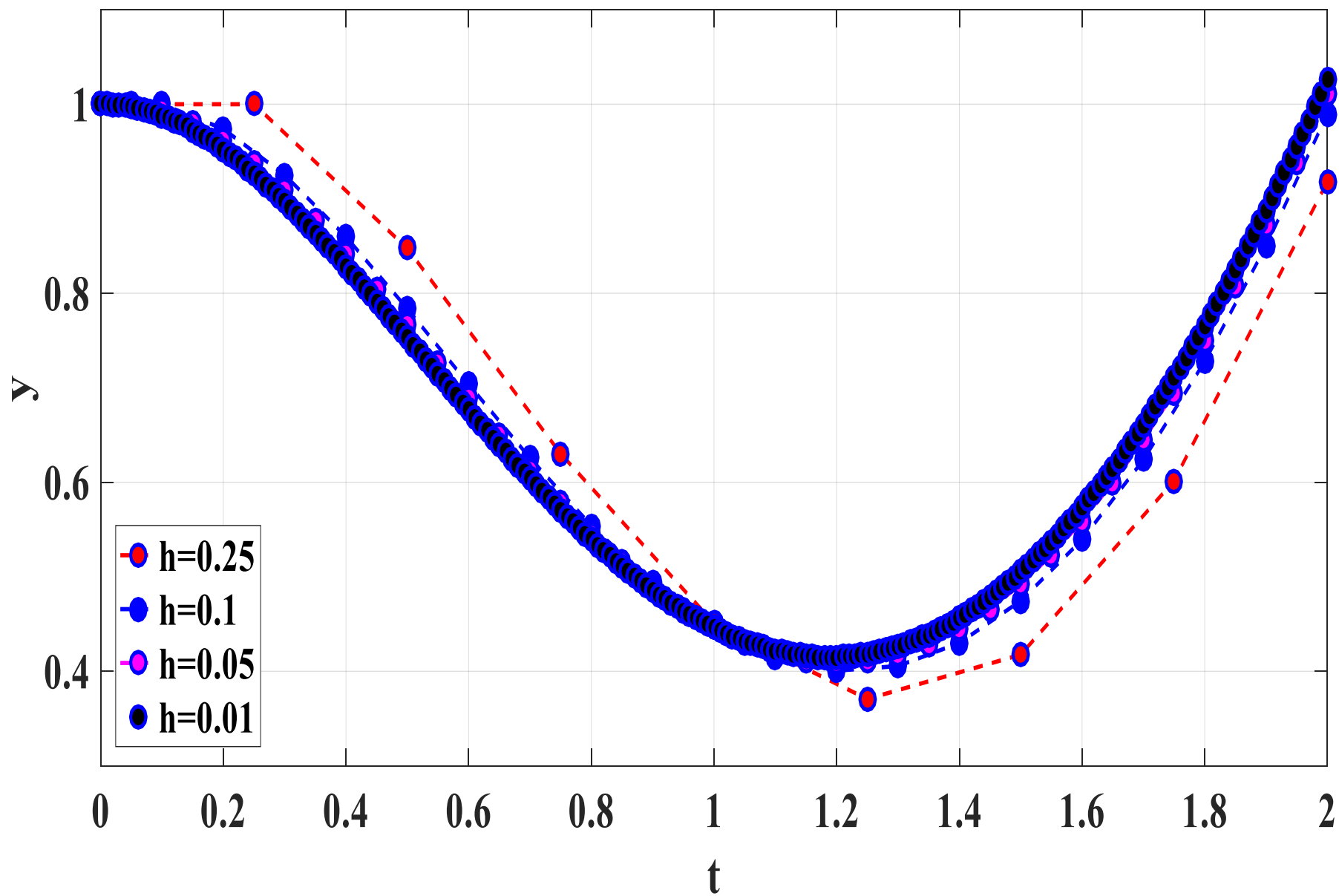
$n=1 \rightarrow x_1 = 1, t_1 = 0.25$

$$\rightarrow x_2 = 1 + 0.25(0.25^2 - e^1(\sin(0.25))) = 0.8475$$

$$\rightarrow x_3 = 0.6303$$

$$\rightarrow x_4 = 0.4508$$

$$\rightarrow x_5 = 0.3706$$



Numerical Solutions of Higher Order DEs

$$y'' = f(t, y, y')$$

Then,

$$x_1(t) = y(t), \quad x_2(t) = y'(t)$$

Therefore:-

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = f(t, x_1, x_2)$$

Example: Second-order system

$$\frac{d^2y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_2 y(t) = bu(t)$$

Define the state variables

$$\begin{aligned} x_1(t) &= y(t) \\ x_2(t) &= \frac{dy(t)}{dt} \end{aligned}$$

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -a_2 x_1 - a_1 x_2 + bu(t)$$

Example: Use Euler's method to find an approximate value for $y(0.3)$, given that

$$y'' - \sqrt{x}y' + e^x y = 1, \quad y(0) = 2 \quad y'(0) = -1 \quad (h = 0.1)$$

Solution: Let $z = y'$ then $z' = y'' = \sqrt{x} z - e^x y + 1$

Now we have Two 1st order ODEs as follows:

$$y' = z \qquad y(0) = 2$$

$$z' = \sqrt{x} z - e^x y + 1 \qquad z(0) = -1$$

According to Euler's method:

$$y(0.1) = y(0) + h z(0) = 2 + 0.1(-1) = 1.9$$

$$\begin{aligned} z(0.1) &= z(0) + h(\sqrt{0} z(0) - e^0 y(0) + 1) \\ &= -1 + 0.1(0 - 2 + 1) = -1.1 \end{aligned}$$

$$y(0.2) = y(0.1) + h z(0.1) = 1.9 + 0.1(-1.1) = 1.79$$

$$\begin{aligned} z(0.2) &= z(0.1) + h(\sqrt{0.1} z(0.1) - e^{0.1} y(0.1) + 1) \\ &= -1.1 + 0.1(\sqrt{0.1} (-1.1) - e^{0.1}(1.9) + 1) = -1.244 \end{aligned}$$

$$y(0.3) = y(0.2) + h z(0.2) = 1.79 + 0.1(-1.244) = 1.6656$$

End of Lecture